

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	ITA0302	Course Title:	Mobile computing
CO Assessed:	CO4	Assessment:	AT2 – CASE STUDY
Weightage:	30 % of CIA	Total Marks:	100
CO4 Statement:	<i>Identify the components and structure of mobile application for Android and Windows OS-based devices.</i>		
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Section / Batch:	Slot D	Date:	27/08/2026

Code of Conduct

I DINDU PUJITHA (192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D PUJITHA

Instructions

- Duration: 60 minutes
- Number of questions : 01

Annexure A - QUESTIONS

1. A mobile application development company wants to create a single application for Android, iOS, and Windows mobile platforms. Developers are debating whether platform-specific frameworks or cross-platform approaches should be used. Analyze the characteristics of the development frameworks available for these operating systems and determine the major trade-offs involved.

Answer:

A company developing one mobile application for **Android, iOS, and Windows** can choose between **platform-specific frameworks** and **cross-platform frameworks**.

Platform-specific development uses separate technologies for each operating system, such as **Kotlin/Java for Android, Swift/Objective-C for iOS, and C#/.NET for Windows**. It provides better performance, platform-specific features, and a native user experience. However, it requires separate codebases, increasing development time, cost, and maintenance effort.

Cross-platform frameworks such as **.NET MAUI, Flutter, and React Native** allow developers to create applications for multiple platforms using a shared codebase. This reduces development time, cost, and maintenance. However, there can be limitations in accessing platform-specific features, achieving completely native performance, and maintaining consistent UI behavior across platforms.

Major Trade-offs:

Factor	Platform-Specific	Cross-Platform
Development cost	High	Lower
Development time	More	Less
Performance	Excellent	Good to Excellent
Code reuse	Low	High
Native features	Excellent	May require extra code
Maintenance	More difficult	Easier
User experience	Highly native	Mostly consistent

Conclusion:

For a company that needs to release the **same application on Android, iOS, and Windows**, a **cross-platform approach is generally more suitable** because it provides high code reuse, faster development, and lower cost. Platform-specific development is preferable when maximum performance and deep integration with each operating system are the main priorities.